Genetic risk, macrophage states and tissue inflammation in rheumatoid arthritis

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Paired RNA/ATAC · Bulk RNA · Spatial transcriptomics · Proteomics

RA synovial tissue, with synovial-fluid and NSCLC comparisons

The scores used in this study

S1 · 15 discovery genes	From sorted active/remission macrophages. Used for exploratory transfer and anti-TNF change.
S2 · 10 predefined IFN genes	Associations in spatial tissue and across bulk disease stages. Only 2 genes overlap with S1.
Hallmark IFN-gamma	Pathway and protein evidence. Specificity is supported in lining, not in sublining.
M-6 / Tph · 30 genes each	Programmes from atlas states, transferred to the fluid and lung cohorts.

Published atlas states and tested integration baselines

31,520 nuclei from 12 donors. RNA and ATAC are measured in the same nuclei.

Fine-state labels, including the 65-nucleus M-6, are taken from Gupta et al.

Concatenation and WNN rank highest in this benchmark. Rare M-6 neighbourhoods favour RNA alone.



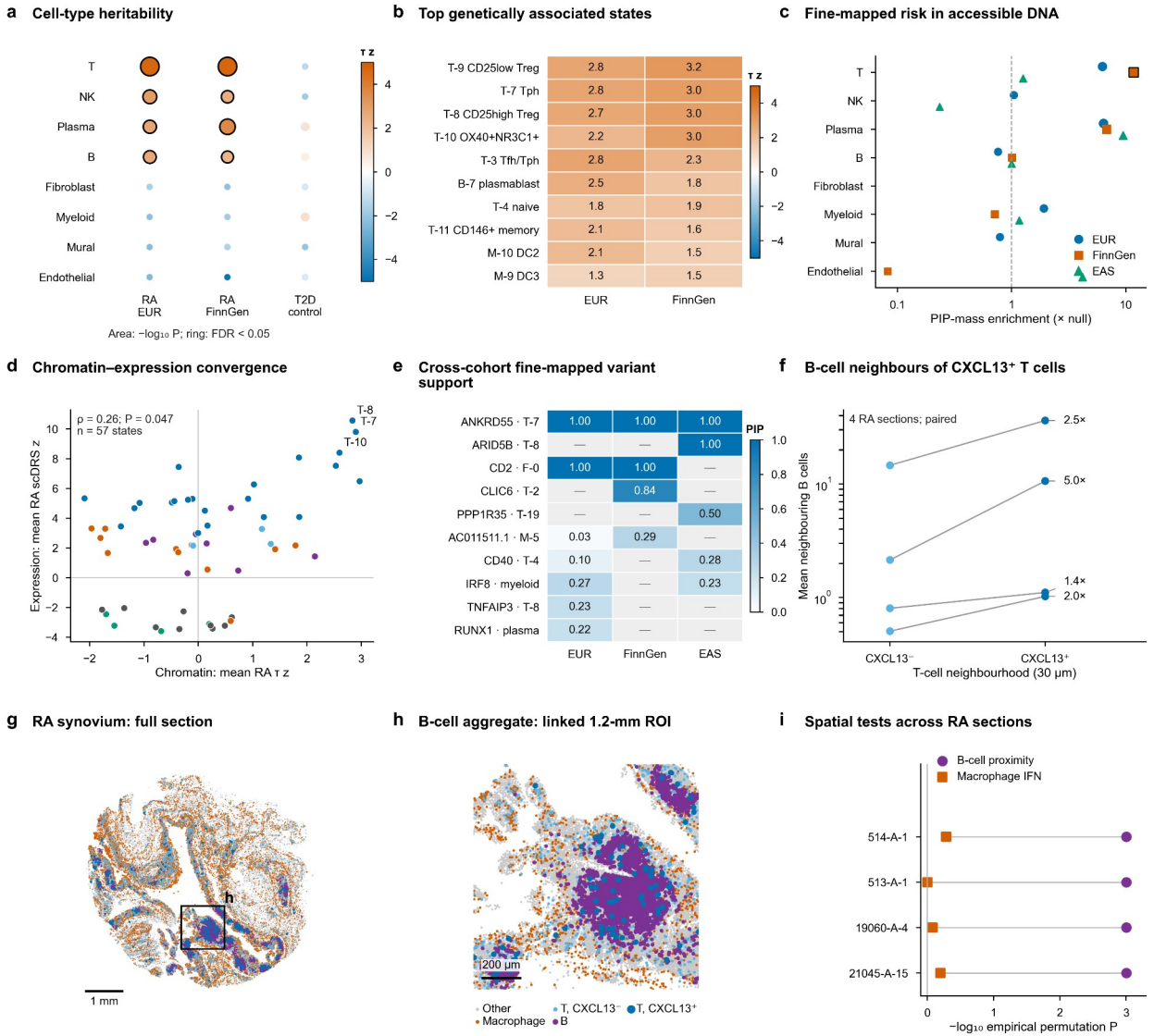
Genetic enrichment and spatial neighbourhood associations

Tph/Treg-associated chromatin is enriched for RA heritability.

S-LDSC and scDRS are complementary methods that use the same GWAS traits.

More B cells sit near CXCL13-positive T cells in 4/4 sections.

The empirical global-permutation P does not account for label clustering or tissue compartments.



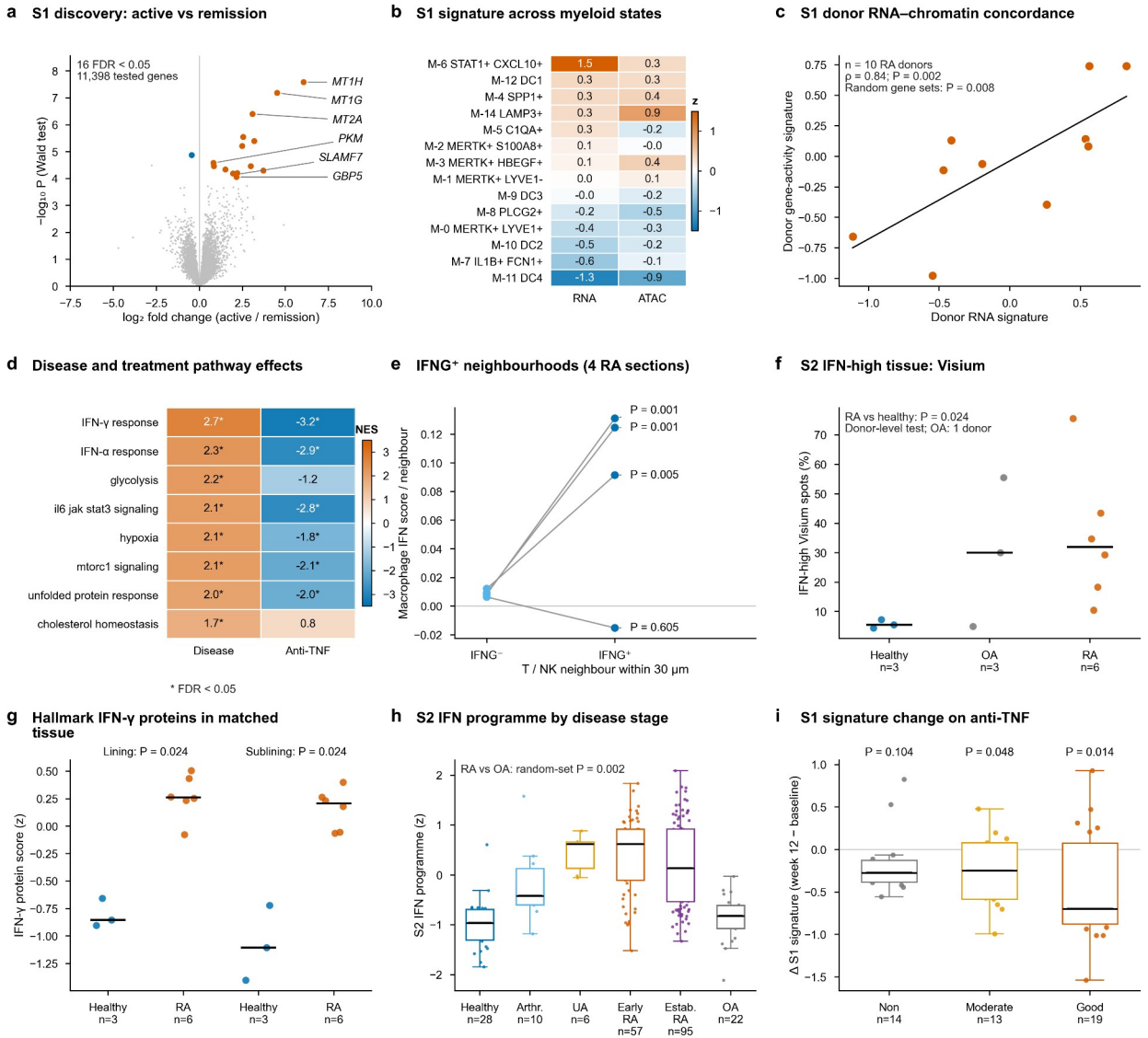
Macrophage signatures across tissue and treatment cohorts

S1 maps to M-6. Its donor inflammation and bulk RA/OA validation tests are negative.

S2 shows spatial and bulk disease-stage associations.

Protein points and test use the same donor averages: 6 RA versus 3 healthy.

S1 falls in good responders, but the change is not significantly different from nonresponders.



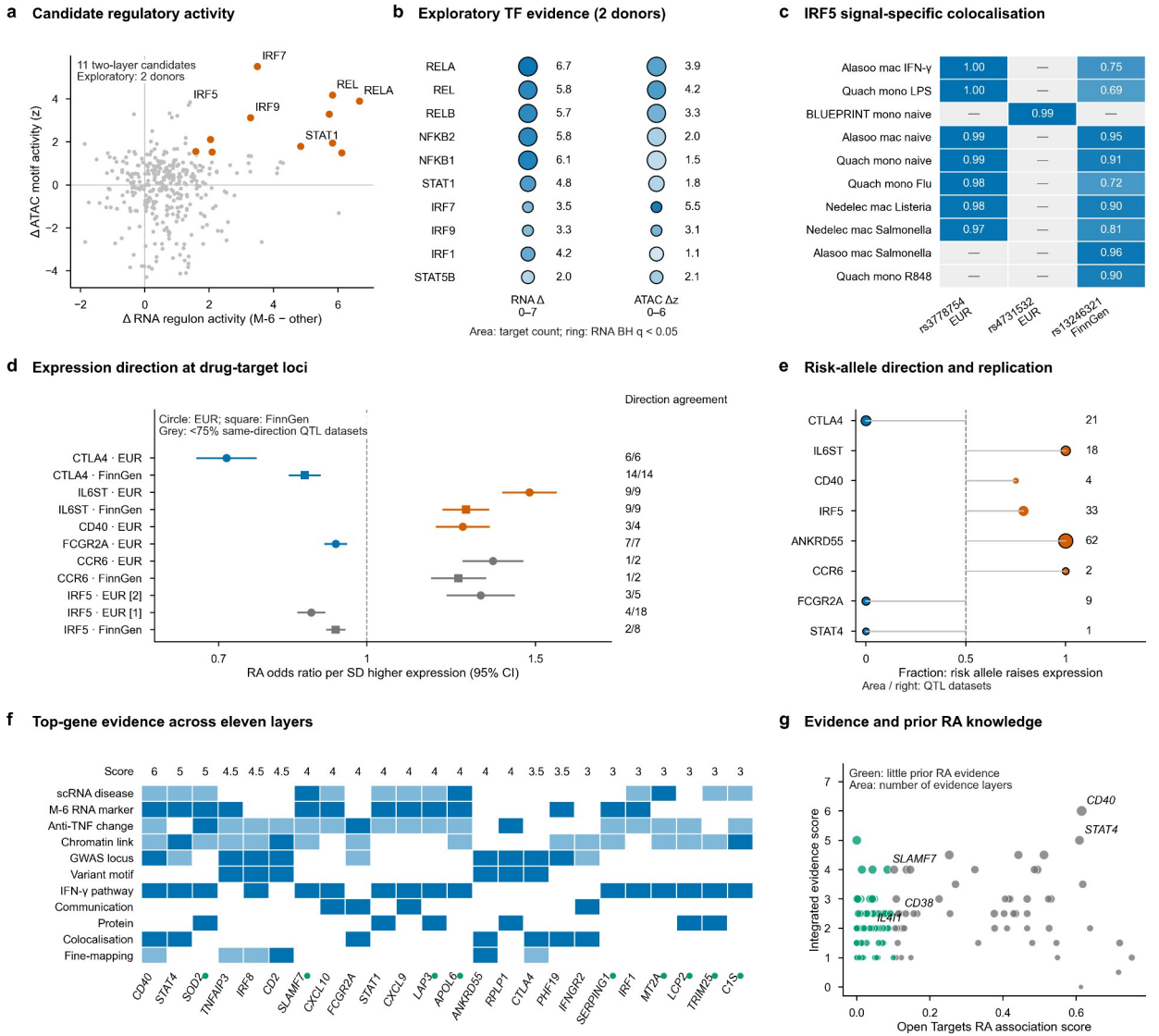
Regulatory candidates and genetic associations

RNA TF tests use BH adjustment across 464 factors. M-6-bearing units come from two donors.

IRF5 colocalisation supports a shared association signal.

Wald ratios cannot estimate drug efficacy or rule out pleiotropy.

Abatacept binds CD80/CD86. IL6ST is not IL6R, the target of tocilizumab.



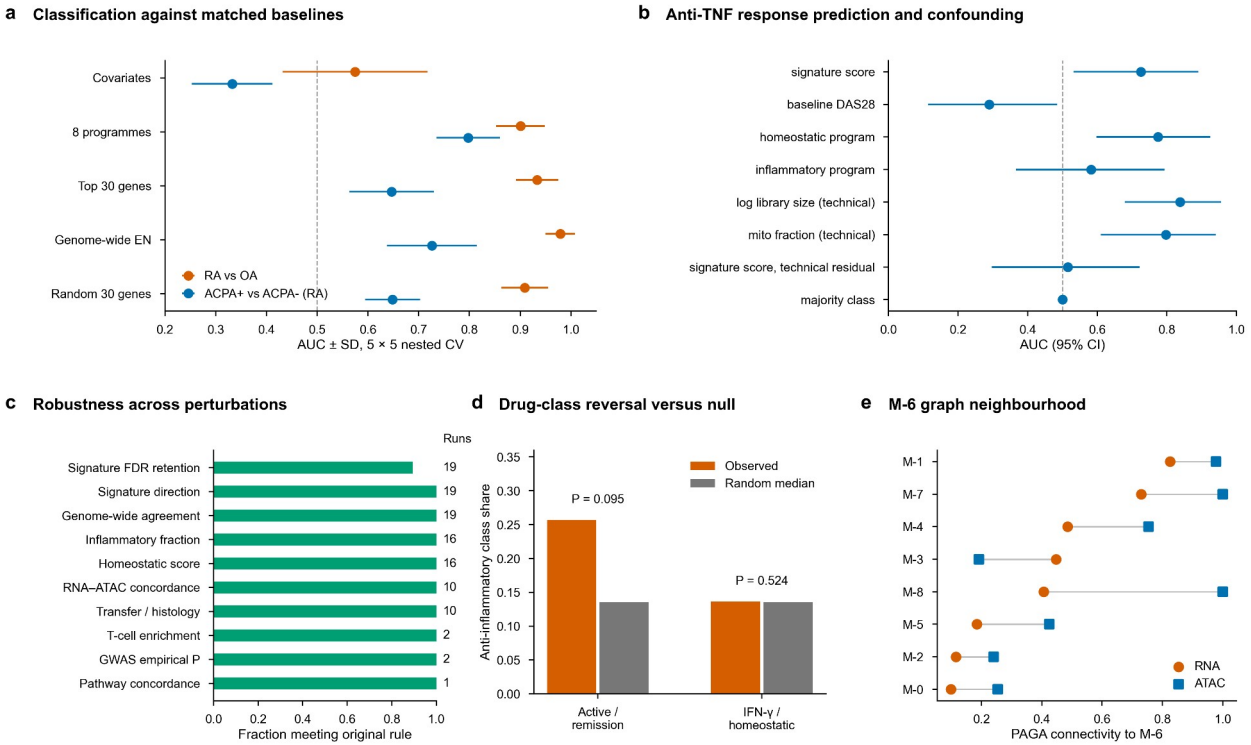
Classification and unsuccessful transfer tests

Classifiers fit all preprocessing on training data only and use repeated nested CV.

199 label permutations repeat the same tuning and resampling as the observed model.

Random controls match the number of retained prioritised genes in each training fold.

Blood transfer, response prediction, deconvolution and trajectory analyses did not support further claims.



f Boundary findings and negative controls

Blood monocytes
Paired signature test: P = 0.45; 13 pairs

Bulk deconvolution
Failed the original sanity checks

Active / remission classification
AUC 0.50–0.60 in the original comparison

RA / OA: prioritised versus random genes
F2 vs random empirical p 0.2475

Bulk Tph score, adjusted for T-cell content
Random-set P = 0.81

Myeloid S-LDSC
Negative conditional τ z in both RA GWAS

Trajectory timing
Chromatin-before-RNA is not interpretable

East Asian T-cell peak enrichment
1.26×; P = 0.22

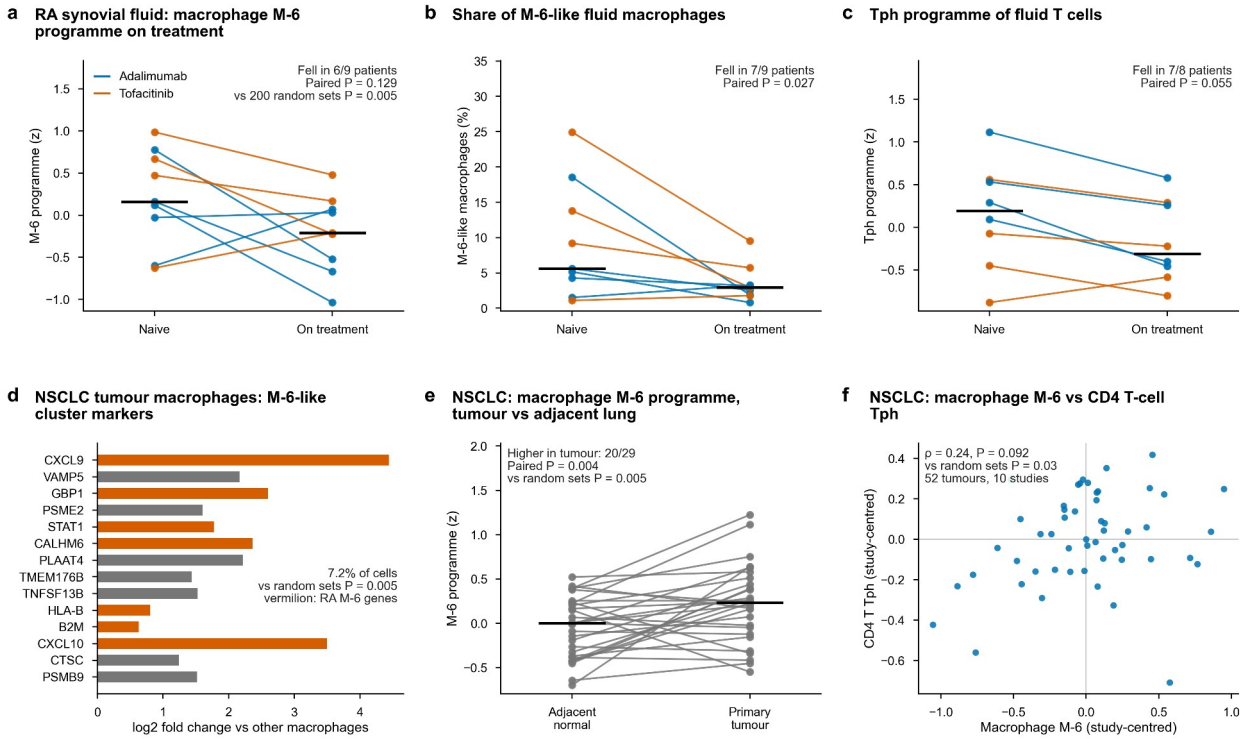
Exploratory transfer to synovial fluid and NSCLC

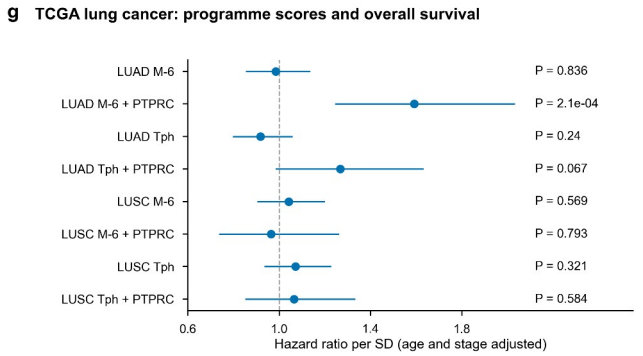
Fluid: 9 paired patients, two therapies pooled. No response labels are available.

The M-6-like fraction falls in 7/9 patients, but the paired pseudobulk-score test is not significant.

LuCA: the M-6 score is higher in tumour in 20/29 paired donors.

Tph coupling P = 0.092. The primary TCGA survival tests are negative.





h Pre-specified tests and verdicts

RA fluid, 9 patients (GSE296117)	
V1 M-6-like cluster exists	borderline (P = 0.055)
V2 M-6 programme falls on treatment	vs random P = 0.005; paired P = 0.129
V3 M-6-like share falls	supported (P = 0.027)
V4 Tph programme falls	borderline (P = 0.055)
NSCLC, LuCA atlas + TCGA	
N1 M-6-like tumour macrophages	supported (P = 0.005)
N2 higher in tumour than adjacent lung	supported (P = 0.004)
N3 couples with CD4 Tph	suggestive (ρ 0.24, P = 0.092)
N4 bulk survival	no association; LUAD HR 1.6 after PTPRC

Classifier performance in nested cross-validation

Mean AUC across 25 outer folds. All fitted transformations stay inside the training folds.
F0 covariates · F1 programmes · F2 prioritised genes · F3 genome-wide

RA vs OA

F0 0.574 ± 0.143

F1 0.900 ± 0.048

F2 0.933 ± 0.042

F3 0.979 ± 0.029

F1 permutation p 0.005

F2 vs random empirical p 0.2475

ACPA+ vs ACPA- (RA)

F0 0.332 ± 0.079

F1 0.797 ± 0.062

F2 0.647 ± 0.083

F3 0.726 ± 0.089

F1 permutation p 0.005

F2 vs random empirical p 0.5149

SD across correlated folds is not a confidence interval. No external clinical validation.

Main findings

RA heritability is enriched in T-cell chromatin; spatial and protein measurements show IFN-associated tissue patterns.

The S1 macrophage signature fails independent inflammation and bulk RA/OA tests.

Prioritised genes do not outperform matched random gene sets in either classification task.

Shared samples and limited donor numbers restrict replication. Mechanisms and clinical utility remain untested.

Reproducing the analyses

Full run: 43 ordered invocations in a clean analysis directory. Environments and external-tool versions are recorded.

The reference snapshot includes API/GMT/eQTL inputs and Open Targets 26.09, with per-file checksums.

The extracted-package check reproduced 112 result tables, 15 processed tables and six figures.

See README.md for installation and run commands; docs/sources.tsv lists accessions and access terms.

Sources and contact

Gupta et al., Nature Genetics 2023: GSE243917 atlas and author-defined states.

Alivernini et al., Nature Medicine 2020: E-MTAB-8322 discovery cohort.

GSE198520 / GSE89408: independent bulk cohorts.

GSE347958 / GSE347797 / PXD062696: shared spatial/protein cohort.

GSE296117 / LuCA / TCGA: exploratory extensions.

All accessions and licences: docs/sources.tsv

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